Personalized Smart Cart with AI Assistance

Simulation

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The Problem

Grocery shopping is often **time-consuming** and **frustrating**:

- Long checkout lines.
- Product information is difficult to read. Challenging for dietary-restricted consumers.
- In-efficient store traversal.
- Forgetting to purchase products leads to repeated visits.

Proposed Solution

A smart cart system with:

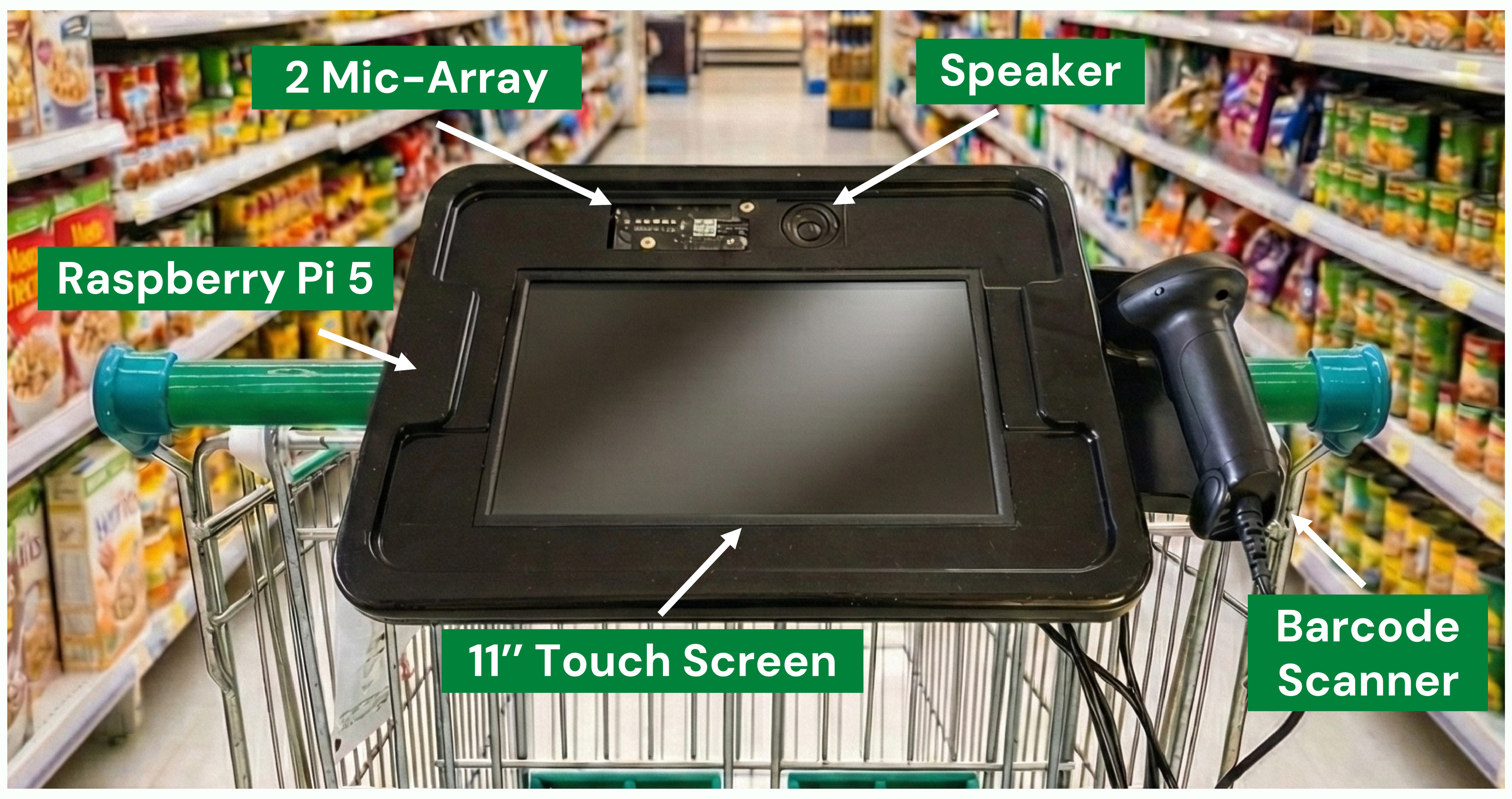
- **AI-based voice assistance** that answers product-related questions, flags dietary conflicts, and suggests suitable alternatives.
- **List-based in-store route** that shows consumers what to pick next, based on their shopping list.
- **Potentially Forgotten Products Suggestions** that recommend products likely due for repurchase and prompt users at checkout.

Essential Requirements

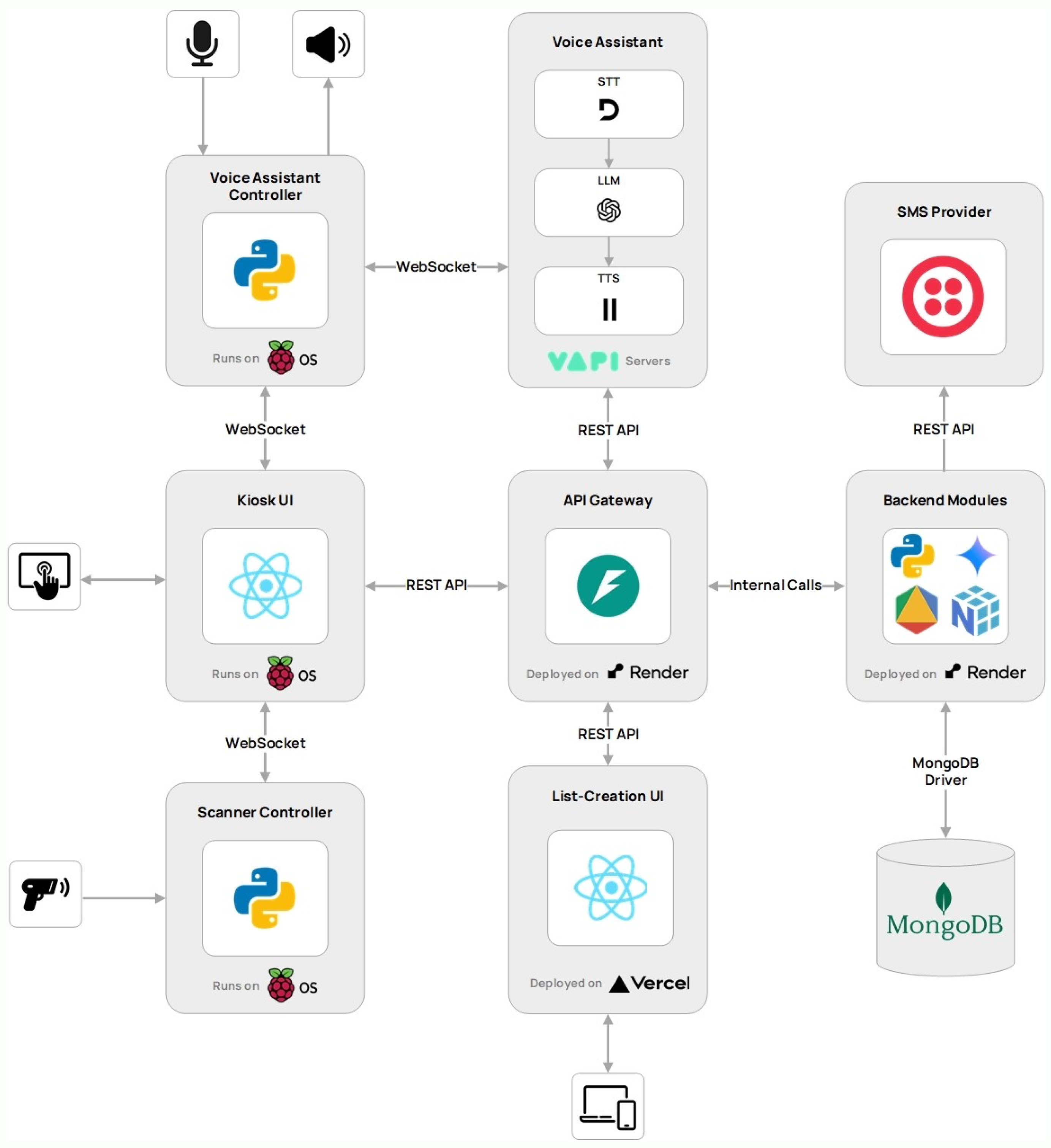
Stability
Concurrent voice-assistant usage by multiple smart carts shall not cause system shutdown.

Performance
System responses shall be provided within reasonable time.

Hardware



System Architecture

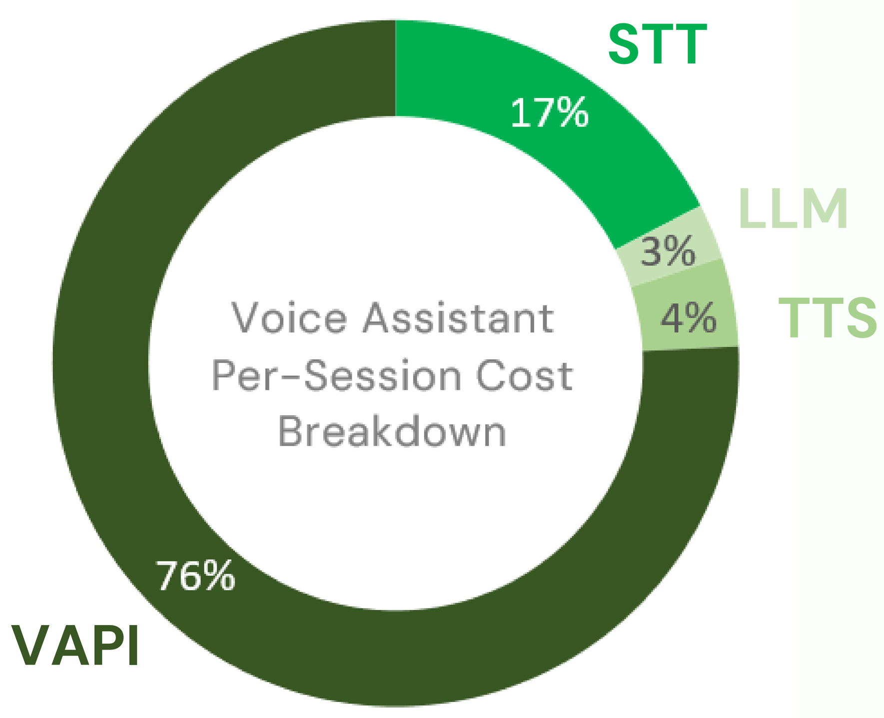


Challenges

Rapidly Evolving Voice-Assistant API
Frequent VAPI API changes led to unexpected failures, required temporary patches and repeated integration updates.

End-to-End System Complexity
Developing a fully functional system within the project timeframe posed a significant challenge for two students.

Integrated Voice Assistant Cost Feasibility



STT, LLM and TTS costs are negligible compared to VAPI.

A custom voice pipeline could significantly reduce costs and enable real deployment.

Evaluation

Aspect	Requirement	Status
Voice Assistant (VA) Latency	< 3 Seconds	✓
Concurrent VA Sessions	≥ 3 Carts	✓
VA Intent-Recognition Accuracy	≥ 90% of Inquiries	94%
SUS	> Average of 68	89.5

Possible Future Enhancements

- Machine learning–based models could enhance frequently-bought product recommendations.
- Recipe-based voice inquires generating optimized route.